WORKSHEET 2.2

The Derivative as a Limit

Unit 2: The Derivative

Key ideas

- $f'(x) = \lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$, wherever this limit exists.
- At a single point: $f'(a) = \lim_{x\to a} \dfrac{f(x) - f(a)}{x - a}$.
- Keep the symbol lim on every line until the limit has actually been taken.
- A limit that has the form of a difference quotient can be **recognised** as a derivative and evaluated with a rule.

Worked examples

Example 1 · A polynomial

Use the definition to find $f'(x)$ for $f(x) = x^2 - 5x$.

Solution. $\lim_{h\to 0} \dfrac{(x+h)^2 - 5(x+h) - x^2 + 5x}{h} = \lim_{h\to 0} \dfrac{2xh + h^2 - 5h}{h} = \lim_{h\to 0}(2x + h - 5)$.

$f'(x) = 2x - 5$.

Example 2 · A square root

Use the definition to find $f'(x)$ for $f(x) = \sqrt{2x+1}$.

Solution. Multiply by the conjugate: $\lim_{h\to 0} \dfrac{(2x+2h+1) - (2x+1)}{h\left(\sqrt{2x+2h+1} + \sqrt{2x+1}\right)} =$

$\lim_{h\to 0} \dfrac{2}{\sqrt{2x+2h+1} + \sqrt{2x+1}}$.

$f'(x) = \dfrac{1}{\sqrt{2x+1}}$.

Example 3 · Recognising a derivative

Evaluate $\lim_{h\to 0} \dfrac{\sqrt{9+h} - 3}{h}$.

Solution. This is $g'(9)$ for $g(x) = \sqrt{x}$, and $g'(x) = \dfrac{1}{2\sqrt{x}}$.

$\dfrac{1}{6}$.

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Practice

CORE

1. Use the definition to find $f'(x)$ for $f(x) = 3x + 2$.

2. Use the definition to find $f'(x)$ for $f(x) = x^2 + x$.

3. Evaluate $\lim_{h \to 0} \dfrac{(5+h)^2 - 25}{h}$.

4. Evaluate $\lim_{x \to 2} \dfrac{x^5 - 32}{x - 2}$ by recognising it as a derivative.

EXAM LEVEL

5. Use the definition to find $f'(x)$ for $f(x) = \dfrac{1}{x^2}$.

6. Use the definition to find $f'(x)$ for $f(x) = x^3 + 2x$.

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7. If $f'(4) = -3$, find $\lim_{h \to 0} \dfrac{f(4+h) - f(4)}{2h}$.

8. Evaluate $\lim_{h \to 0} \dfrac{\ln(2+h) - \ln 2}{h}$.

Question 9 · Free-response practice

Let $f(x) = \dfrac{3}{x-1}$ for $x \neq 1$.

(a) Use the limit definition to find $f'(x)$.

(b) Write an equation of the line tangent to the graph of f at $x = 2$.

(c) Evaluate $\lim_{h \to 0} \dfrac{f(4+h) - f(4)}{h}$.

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Worked solutions

1. $\lim\limits_{h\to 0}\dfrac{3(x+h)+2-3x-2}{h}=\lim\limits_{h\to 0}3=3.$

2. $\lim\limits_{h\to 0}\dfrac{2xh+h^2+h}{h}=2x+1.$

3. The derivative of x^2 at 5: 10.

4. The derivative of x^5 at 2: $5(2)^4=80$.

5. $\lim\limits_{h\to 0}\dfrac{x^2-(x+h)^2}{h\,x^2(x+h)^2}=\lim\limits_{h\to 0}\dfrac{-2x-h}{x^2(x+h)^2}=-\dfrac{2}{x^3}.$

6. $\lim\limits_{h\to 0}\dfrac{3x^2h+3xh^2+h^3+2h}{h}=3x^2+2.$

7. $\dfrac{1}{2}f'(4)=-\dfrac{3}{2}.$

8. The derivative of $\ln x$ at 2: $\dfrac{1}{2}$.

9. (a) $\lim\limits_{h\to 0}\dfrac{\frac{3}{x+h-1}-\frac{3}{x-1}}{h}=\lim\limits_{h\to 0}\dfrac{-3}{(x+h-1)(x-1)}=-\dfrac{3}{(x-1)^2}.$

   (b) $f(2)=3$ and $f'(2)=-3$: $y-3=-3(x-2)$.

   (c) This is $f'(4)=-\dfrac{3}{9}=-\dfrac{1}{3}$.

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